Analytical Chemistry Cumulative Exam

October 28, 2022

This Cumulative Exam focuses on basic undergraduate analytical chemistry.
(At the end of this cumulative exam are equations that might or might not be helpful in answering these questions)

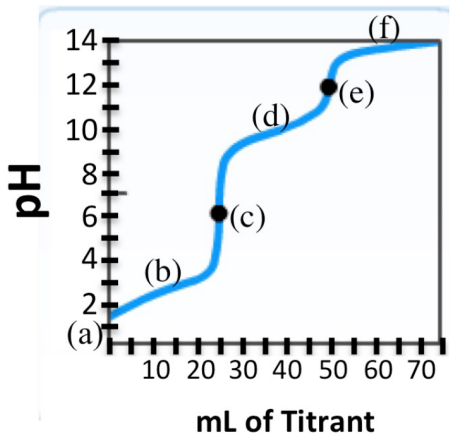
1. (15 points) You titrate 60.0 mL of 0.1200 M $Pb(NO_3)_2$ with 140.0 mL of 0.0420 M NaI. Calculate the pPb^{2+}. The K_{sp} for $PbI_{2(s)}$ is 7.9×10^{-9}. ***(Show work)***

2. (15 points total) A weak acid, H_2A, (0.100 M) is titrated with a strong base. Use the figure here to answer the following questions. (The y-axis ranges from pH = 0 to 14.)

(a) (3 points) What ***letter*** in the figure best represents the location of the second equivalence point? Explain why?

(b) (5 points) Estimate the ***value*** of pK_{a1} (not the letter) from the figure where $[H_2A] / [HA^-] = 1$.

(c) (7 points) Estimate the ***value*** of K_{a2} from the figure. *(Show work.)*



3. (40 points) A solution of **25.00 mL** of **0.1000 M** disodium maleate NaOOC-CH=CH-COONa (the deprotonated form of maleate, which can be represented by A^{2-}) is titrated with a strong acid. Note that the two K_a's of the diprotic form of maleate (maleic acid) are: $\mathbf{K_{a1} = 1.3 \times 10^{-2}, K_{a2} = 5.9 \times 10^{-7}}$.
Answer all three of the following questions for this titration.

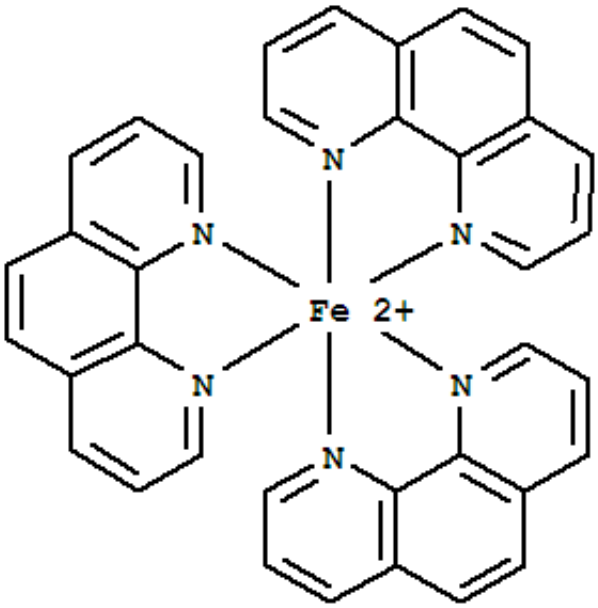
(a) **(10 points)** What is the pH of the solution before the titration begins (**0.1000 M of A^{2-}**)? (show work)

(b) **(15 points)** What is the pH of the solution ***after*** adding **50.00 mL** of **0.05000 M HCl**? (show work)

(c) **(15 points)** What is the pH of the solution ***after*** adding **150.00 mL** of **0.0500 M HCl**? (show work)

4. (15 points) Explain the difference between sensitivity and detection limit. Use equations and diagrams to support your discussion both quantitatively and visually.

5. (15 points) Name two different instrumental methods that you can use to ***quantify*** the compound shown below, Tris(1,10-phenanthroline)iron(II), in a sample. Explain why each method would be appropriate. Note that this analyte is in a mixture with other compounds that arrives to your laboratory in a solid form. Describe the sample preparation procedures that you would perform, if any, before the instrumental analysis. Note any assumptions that you would have to make, if any, in your analysis approach.



http://www.guidechem.com/reference/dic-329192.html

Equations that might or might not be relevant to acid-base equilibria

(for Analytical Chemistry Cume)
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$$[H_3O^+]^3 + K_a\,[H_3O^+]^2 - \left(K_w + K_a\,C_{HA}\right)[H_3O^+] - K_a\,K_w = 0$$

$$[H_3O^+] \approx \frac{-K_a + \sqrt{K_a^2 + 4\,K_a\,C_{HA}}}{2}$$

$$[H_3O^+] \approx \sqrt{K_a\,C_{HA}}$$

$$[OH^-]^3 + K_b\,[OH^-]^2 - \left(K_w + K_b\,C_{NaA}\right)[OH^-] - K_b\,K_w = 0$$

$$[OH^-] \approx \frac{-K_b + \sqrt{K_b^2 + 4\,K_b\,C_{NaA}}}{2}$$

$$[OH^-] \approx \sqrt{K_b\,C_{NaA}}$$

$$[H_3O^+] = \sqrt{\frac{K_{a_2}\,[HA^-] + K_w}{1 + [HA^-]/K_{a_1}}}$$

$$[H_3O^+] = \sqrt{\frac{K_{a_2}\,C_{NaHA} + K_w}{1 + C_{NaHA}/K_{a_1}}}$$

$$[H_3O^+] \approx \sqrt{K_{a_1}\,K_{a_2}}$$

$$pH = -\log[H_3O^+] \approx pKa + \log\left(\frac{C_{NaA}}{C_{HA}}\right)$$

PERIODIC TABLE OF ELEMENTS

Atomic # / Symbol / Name / Weight

- C Solid
- Hg Liquid
- H Gas
- Rf Unknown

Metals: Alkali metals; Alkaline earth metals; Lanthanoids (Lanthanides); Actinoids (Actinides); Transition metals; Post-transition metals

Metalloids

Nonmetals: Other nonmetals; Noble gases

Pnictogens; Chalcogens; Halogens

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.008																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

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